

XRComm FCS 8T8R Platform

Application Development Guide

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Platform	XRComm FCS 8T8R Platform
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Introduction

This document provides detailed guidelines for developers writing, compiling, and running custom applications that interface with the XRComm FCS 8T8R Platform.

The platform provides a header-only, zero-copy C API (`xrcomm_ip_client.h`) for POSIX applications that consume I/Q sample batches.

Compiling and Running the Hello World Examples

The platform includes an example consumer application in the `XRComm_8T8R_hello_world/` folder that demonstrates how to subscribe to and consume platform data streams.

Example Binary	Delivery Mode	Enable With	Privileges
hello_world	IQ samples (zero-copy shared memory)	set-ip on	None

Build the Hello-World Application

Navigate to the hello world folder:

```
cd XRComm_8T8R_hello_world/
```

Compile using CMake:

```
mkdir build && cd build
cmake .. -DXRCOMM_INCLUDE_DIR=../XRComm_platform_drivers/include
cmake --build .
```

Or compile directly with GCC:

```
gcc -O2 -o hello_world hello_world.c \
-I../XRComm_platform_drivers/include \
-lrt -lpthread -lm
```

Compiled binaries are placed in `XRComm_8T8R_hello_world/bin/` or in the root of the hello world folder.

Run the Hello-World Application

The platform driver must be running and IQ mode must be enabled before starting the application:

Terminal 1 (FlexServer): Ensure `xrcomm_server` is running, then enable IQ mode:

```
python3 XRComm_platform_gRPC/client/xrcomm_grpc_client.py set-ip on
```

Terminal 2 (FlexServer): Run the application:

```
./hello_world
```

The example prints a summary of batches, samples, signal power, and drops for all 8 channels:

```
[IP:hello_world_ch0] Registered slot 0 (pid 12345, port 0, ch 0)
...
[HW] All channels ready – starting receive loop
[HW] Channel summary (port 0):
  ch0: batches=15234 samples=487488 power=-18.4dBFS mean=-18.6dBFS dropped=0
  ch1: batches=15234 samples=487488 power=-19.1dBFS mean=-19.2dBFS dropped=0
  ...
```

Verify the Application Is Registered

Query the registry from the **Client Computer**:

```
python3 xrcomm_grpc_client.py --host <server-ip>:50051 list-ips
```

Each of the 8 channel slots should appear active:

```
slot=0  name=hello_world_ch0  pid=12345  mode=IQ_MODE  port=0  ch=0  ring=ready  active=yes
slot=1  name=hello_world_ch1  pid=12345  mode=IQ_MODE  port=0  ch=1  ring=ready  active=yes
...
```

Application API Reference

Include `xrcomm_ip_client.h` (found in `XRComm_platform_drivers/include/`) in your applications. No DPDK headers are required.

Connection & Lifecycle

xrcomm_ip_client_init_port_channel

```
int xrcomm_ip_client_init_port_channel(xrcomm_ip_client_t *client,
                                     const char *name,
                                     uint32_t hsp_port,
                                     uint32_t channel);
```

8T8R: Registers the application to consume the IQ stream of a single channel of one HSP port. Arguments: client handle, application name (max 63 chars), hsp_port (0 or 1), channel (0–7). Returns 0 on success.

xrcomm_ip_client_wait_ready

```
int xrcomm_ip_client_wait_ready(xrcomm_ip_client_t *client, int timeout_ms);
```

Blocks the caller until the platform has created and opened the shared memory data ring for this channel. Timeout of 0 means wait indefinitely. Returns 0 when ready, -1 on timeout.

xrcomm_ip_client_running

```
bool xrcomm_ip_client_running(xrcomm_ip_client_t *client);
```

Returns true as long as the platform's receive loop is active. Use this as the main condition for your ingest loop.

xrcomm_ip_client_shutdown

```
void xrcomm_ip_client_shutdown(xrcomm_ip_client_t *client);
```

Deregisters the channel handle from the platform and releases all mapped shared memory regions.

Data Consumer Functions

xrcomm_ip_client_poll

```
const xrcomm_iq_slot_t* xrcomm_ip_client_poll(xrcomm_ip_client_t *client);
```

Polls the data ring for a new batch of IQ samples. Returns a read-only pointer to the batch if available, or NULL if empty or if IQ mode is off. Batch fields: - `samples[]`: Array of interleaved little-endian int16 IQ pairs - `valid_sample_count`: Count of valid samples in the batch - `hsp_port`, `channel`: Metadata indicating source port and channel

xrcomm_ip_client_consume

```
void xrcomm_ip_client_consume(xrcomm_ip_client_t *client);
```

Releases the current batch slot and returns it to the platform's write pool. Must be called once per successful poll.

Diagnostics

xrcomm_ip_client_get_dropped

```
uint64_t xrcomm_ip_client_get_dropped(xrcomm_ip_client_t *client);
```

Returns the number of batches dropped by the platform for this subscription because the application polled too slowly.

xrcomm_ip_client_report_spectrum

```
void xrcomm_ip_client_report_spectrum(xrcomm_ip_client_t *client,  
                                     uint64_t processed,  
                                     float power_dbfs);
```

Reports processing stats back to the platform, making them visible in gRPC telemetry and statistics.